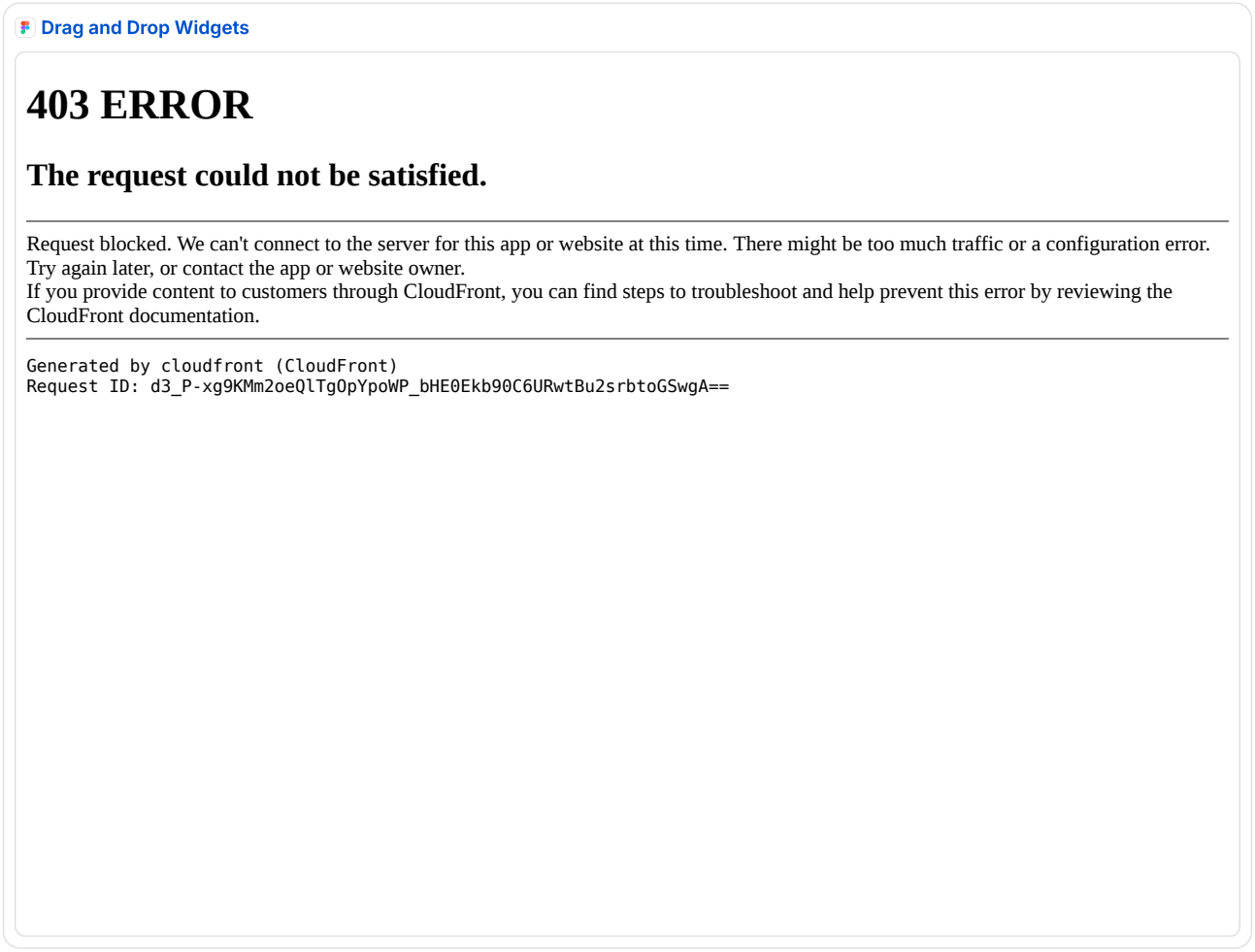
Sorter

The Sorter widget allows learners to arrange items in a specified order. This widget is ideal for sorting related content items along a spectrum scale between two categories, placing events on a timeline, or crafting a complete sentence.



Figma Make Prototype

Anatomy

- Answer Zone
  - Legend
  - Blank(s)
- Choice Bank
  - Answer Tiles

Behavior

This widget follows the Overview document except where noted below.

Variants

The Sorter widget offers three distinct variants to support specific exercise types: Scale, Timeline, and Sentence.

Scale

The Scale variant allows learners to arrange items along a spectrum scale between two categories.

Timeline

The Timeline variant allows learners to arrange events in chronological order between two specified dates.

Sentence

The Sentence variant allows learners to arrange words and punctuation in order to form a complete sentence.

Answer Zone

The order widget has different configurations for the answer zone depending on the widget variant.

Legend

The widget has a legend with descriptive visible labels so that a learner knows the order in which they are expected to place the answer tiles. This widget does not require a legend — however, exercises without labeled legends are marked as inaccessible and excluded from the practice experience for the learners that opt-out of inaccessible learning content. Legends and labels are configured in the content editor.

Scale and timeline variants require two labels, one placed at the start and the other placed at the end of the legend. These labels bound the scale or timeframe for the exercise. Custom screen reader labels are required for the legend so that the scale or timeline is adequately textually conveyed where the visual border is the sole method of communication. For example: interest rates ranked from lowest at the start to highest at the end.

Legends must have a defining border to distinguish them from other content and convey the organizational axis expected for the answer tiles.

Legends for the scale and timeline variants are configurable so that they can appear horizontally or vertically, at the start or end of the blanks where the answer tiles are placed. Horizontal orientations allow the legend to display above or below the blanks. Vertical orientations only support a legend at the start of the answer zone and not the end. All legend configurations adapt their orientation to the vertical orientation when the width of the parent container is reduced to where the space is insufficient to preserve the operability and understanding of the content without the content being cut off. The Timeline variant allows the legend above the blanks when the number of blanks is less than 5.

The start and end style of the legend border can also be configured for the scale and timeline variants. Each end is separately configurable, with the following style choices:

- Arrow (best for scale)
- Bar (best for timeline)
- None

When arrows and bars are selected for the border, their orientation should adjust with the border itself so that the arrows point up or down and the bars are horizontal instead of vertical.

Legend position and appearance for the sentence variant are not configurable as they are for the other widget variants. Sentence variant labels display above the blank and are justified along the start axis of the user's language's writing direction.

Blanks

Blank counts and sizing vary based on the widget variant. These specifications are determined by what works best for the display and responsive behavior based on varying parent container widths.

The Sentence variant has a single blank that behaves identically to the blank in the Composer widget, where it is both larger and wider than the answer tiles it accepts, filling the width and height of its column parent container.

The Scale variant allows 2 - 10 blanks. When the widget has 2 - 4 blanks, it can display in the horizontal orientation. When the widget has 5 - 10 blanks, it displays in the vertical orientation only, regardless of viewport size. Widgets with 5 - 10 blanks are discouraged.

Blanks are evenly spaced apart, with each blank along the legend filling the available space. Blank height matches answer tile height. When the width of the parent container narrows so that the blank size decreases to the point where they are not wide enough to house the answer tiles, the orientation of the widget must change to the vertical orientation where the blanks stack vertically and take up all of the viewport width available after the legend.

The Timeline variant allows 3 - 7 blanks. Blanks display above the legend first and are evenly spaced apart, with each blank along the legend filling the available space. Timeline variant blanks have lines associated with them, centered in the width of the blank, that connect it to the legend and allow the user to understand the order of the blank based on its connection and orientation to the legend. Blank height matches answer tile height.

When the number of blanks exceeds 4, the blanks are split above and below the legend, staggered in a centered, alternating tree arrangement until they reach the maximum of 7. Odd number blanks display above the legend, even number blanks below it. When the number of blanks is 5, three display above the legend and two display beneath it. When the number is 6, three display above and below the legend. When number is 7, 4 display above and 3 display below.

If a horizontally oriented timeline variant has 4 or fewer blanks, the blanks should stagger above and below the legend as the available parent container width decreases. The stagger pattern should alternate, placing odd blanks above and even blanks below the legend border.

Blanks should fill the width available and scale down until they cannot house the full answer tile, at which point the widget should switch to the vertical orientation with the Legend at the start: placed on the left with blanks on the right in Left-to-Right writing directions and placed on the right with blanks on the left in Right-to-Left writing directions.

Blanks fill 100% of the container width available between the legend and viewport end, and their connector lines display at the start side of the blank, connecting them to the legend in linear (non-staggered) order.

Scale and timeline blanks are numbered based on their order along the legend. Variants with multiple blanks in the legend area should be contained in an ordered list where the ol is labeled by the legend text.

Answer Tiles

Answer tiles in the Sorter Widget do not allow an empty content type. They allow all other content types.

Scale and Timeline Variants

- Allow single-use tiles only, as organizing tiles in these scenarios is a mutually exclusive exercise.
- Answer tiles fill the size of the containing blank when placed.

- The actions menu remains visible when a tile is placed in a blank.
- When an answer tile is dragged or moved via the actions menu to a blank with another answer tile already placed in it, the new tile replaces the old tile and the old tile is moved back to the choice bank.

Sentence Variant

- Allows both single and multi-use answer tiles in the choices bank, but they cannot be mixed and matched. All tiles must be the same use type so that users are not confused when seemingly identical controls behave in divergent ways.
- Answer tiles hug their contents when placed.
- The actions menu button displays only on hover or focus when a tile is placed in a blank.

Actions Menu

Sentence Variant

When an answer tile is in the choices bank, the actions available allow a learner to place a tile at the start of the blank. After an answer tile is placed in a blank, the menu actions adapt to include only what's possible for that specific tile based on its placement. Features that are available to mouse and touch users must be available to keyboard and screen reader users, so the actions menu includes multiple potential actions for a given tile: Move up, down, to start, to end, or Clear.

The start and end tiles have a subset of available actions since they are unable to move in both directions. Start tiles show move down, to end, and clear as the available actions. End tiles show move up, to start, and clear as the available actions.

Configuration

The Sorter widget requires the answer tiles to be in a specific order to be scored correct. Each answer tile is assigned a numeric value in the content editor. For the Scale and Timeline variants, the answer tile value must match the blank's number (positional key value pairs) within its ordered list to be scored correct. For the Sentence variant, when the sequence of tiles in the blank matches the correct incrementing sequence, the learner is scored correct.

Best Practices

- Legend text should not exceed one line when displayed at their maximum width. Character limit warnings should display in the editor to help guide content authors.
- Answer tiles should be succinct, eliminating all unnecessary words, especially for the Sentence and Timeline variants.